11501_FinCloud_13_MLOps｜MVP 截圖與重現步驟

```
$ git clone <repo> && cd fincloud-mlops-risk && pip install -r requirements.txt
$ make all
python data/generate_data.py --n 10000 --seed 42 --out data/raw/applications.csv
wrote 10100 rows -> data/raw/applications.csv
python data/generate_data.py --n 3000 --seed 7 --drift 1.0 --out data/raw/applications_2026q1.csv
wrote 3030 rows -> data/raw/applications_2026q1.csv
python -m src.clean
{"rows": 10100, "duplicates": 100, "missing_income": 300, "invalid_debt_ratio": 55, "default_rate":
0.2158, "rows_after_clean": 10000}
python -m src.train
{"logistic": {"auc": 0.7638, "ks": 0.397}, "gbm": {"auc": 0.7411, "ks": 0.3686}, "n_train": 7500,
"n_test": 2500}
python -m src.monitor data/raw/applications.csv data/raw/applications_2026q1.csv
{"ALERT": ["debt_ratio", "credit_utilization", "late_payments_12m"], "retrain": true}
$ python -m pytest -q
.....                                                                    [100%]
5 passed in 0.32s
$ python src/local_demo.py   # 模擬 Lambda 評分
{"model_version": "v20261104150738", "pd": 0.7957, "decision": "MANUAL_REVIEW_HIGH_RISK", "top_factors":
["late_payments_12m", "credit_utilization", "debt_ratio"]}
{"model_version": "v20261104150738", "pd": 0.0131, "decision": "AUTO_APPROVE", "top_factors":
["late_payments_12m", "credit_utilization", "debt_ratio"]}
$ cat models/registry.json | tail -1 (節錄)
{"version": "v20261104150738", "champion": "logistic", "status": "staging", "sha256": "c79daa00a4ff"}
```

圖 1　全新 clone 後執行的實際輸出（資料筆數 10,100 → 10,000；AUC 0.764；5 項測試通過）

重現步驟

1. `git clone https://github.com/chinglin-k/fincloud-mlops-risk && cd fincloud-mlops-risk`
2. `pip install -r requirements.txt`
3. `make all`：產生合成資料（seed=42）、清理分層、訓練並寫入模型登錄、漂移監控
4. `python -m pytest -q`：5 項測試通過
5. Learner Lab：Start Lab → 憑證貼到 ~/.aws/credentials → `bash aws/deploy_learner_lab.sh`（建立加密 S3、上傳 raw／processed／curated 與模型、Athena 建表查詢、DynamoDB 稽核表、Lambda risk-score 並呼叫一次）
6. 驗證：Athena `SELECT COUNT(*) FROM fincloud_risk.features` = 10,000；Lambda 回傳 PD 與 decision；DynamoDB risk_score_audit 有一筆紀錄
7. 完成後 End Lab

設定：Region us-east-1；角色 LabRole；Lambda python3.12、timeout 10 秒；S3 SSE-S3 加密、封鎖公開存取、版本控制。

注意：Learner Lab 主控臺截圖（S3、Athena 查詢結果、Lambda 測試、DynamoDB 項目）請組員執行步驟 5 後補上。